INCH-POUND

MIL-I-44415
21 December 1990

MILITARY SPECIFICATION

INSECT REPELLENT, PERSONAL APPLICATION (CREAM LOTION, TUBE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of extended duration insect repellent in cream lotion form in a plastic tube for personal application to the skin.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6840

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MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act (40 CFR Part 162)

(Application for copies should be addressed to the Environmental Protection Agency, 401 M Street SW, Washington, DC 20460.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).



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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1639 - Acid Value of Organic Coating Materials, Test Method For

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP)

(Application for copies should be addressed to the United States Pharmacopeia Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 N, N-diethyl-m-toluamide (DEET). The DEET used in formulating the insect repellent shall be technical grade and shall contain not less than 95 percent purity by weight of meta isomer. The remainder shall consist almost entirely of ortho or para isomers of DEET or a mixture of both isomers as indicated by the infrared absorption spectra or gas chromatograph.

3.2 Composition. The insect repellent shall be registered with the U.S. Environmental Protection Agency (EPA) in accordance with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA). The insect repellent shall have the chemical composition in table I and meet the requirements in table II.

TABLE I. Insect repellent composition (see 6.4)

| Chemical | Percent by weight |
|---|---|
| Purified water USP | 41.54 |
| Mixture of 25% IP-III polymer in DEET | 23.33 |



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TABLE I. Insect repellent composition (see 6.4) (cont'd)

| Chemical | Percent by weight |
|---|---|
| DEET | 17.50 |
| Propylene glycol dicaprylate/dicaprylate | 2.41 |
| Polyethylene glycol (7) glyceryl ether | 2.26 |
| Propylene glycol (15) stearyl ether | 2.20 |
| Cetyl stearyl alcohol | 2.20 |
| Polyethylene glycol (82) glyceryl monotallowate | 1.65 |
| Amorphous silicon dioxide | 1.64 |
| Glyceryl stearate/sodium lauryl sulfate | 1.55 |
| Polyethylene glycol (8) | 0.98 |
| Magnesium aluminum silicate | 0.70 |
| Polyethylene glycol (200) glyceryl monotallowate | 0.65 |
| Cetyl palmitate | 0.65 |
| Cellulose, 2-hydroxyethyl ether | 0.50 |
| Diazolidinyl: urea: methyl paraben: propyl paraben: propylene glycol | 0.24 |

TABLE II. Chemical and physical properties

| Properties | Requirement |
|---|---|
| DEET content | 31.58 min./36.75 percent max. |
| Other (DEET) isomers content | 1.75 percent max. |



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TABLE II. Chemical and physical properties (cont'd)

| Properties | Requirement |
|---|---|
| Viscosity | 150,000 - 250,000 cps (LVT #TE @ 0.6 rpm) |
| Acid number | 22,000 - 36,000 cps (RVT, #TB @ 5 rpm) 0.15 maximum |
| Bacteria plate count | Not more than 100 CFU/mL, no pathogens |
| Density | 0.990 - 1.035 g/mL |
| Odor | Near odorless, except for slight characteristic odor of diethyltoluamide |
| pH | 6.9 - 8.0 |
| Color | White viscous cream |

3.3 Shelf life. Each production lot shall remain stable with DEET content not less than 31 percent for at least 5 years.

3.4 Workmanship. The manufacturer shall implement procedures and controls to ensure that the process and the product produced are not compromised by foreign materials and contaminants or any other conditions which may degrade the composition. The material shall be smooth, free of lumps, and homogeneous when examined visually or by touch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.



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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed in table III conform to the specified requirements when tested by the specified methods.

TABLE III. Component and material tests

| Characteristic | Requirement reference | Test method |
|---|---|---|
| DEET content | Table II | 4.3.1 |
| Other (DEET) isomers content | Table II | 4.3.1 |
| Viscosity | Table II | 4.3.4 |
| Acid number | Table II | 4.3.3 |
| Bacteria plate count | Table II | 4.3.7 |
| Density | Table II | 4.3.2 |
| Odor | Table II | 4.3.6 |
| pH | Table II | 4.3.8 |



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4.2.2 End item visual examination. The end items (filled, sealed, and labeled tubes of insect repellent) shall be examined for the defects listed in table IV. The lot size shall be expressed in units of tubes. The sample unit shall be one tube. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5.

TABLE IV. End item visual defects

| Examine | Defect |
|---|---|
| Construction | Not as specified (see 5.1.1) |
| Closure | Not as specified (see 5.1.1) |
| Markings (label) | Missing, incomplete, incorrect, illegible, not as specified (see 5.4) |
| Workmanship | Dirty; any leaking of contents; any other defect affecting serviceability |

4.2.3 End item contents examination. The end items (filled, sealed, and labeled tubes of insect repellent) shall be examined for net contents in accordance with 4.3.5 and for presence of foreign particles. After weighing of the tubes in accordance with 4.3.5, the contents of the tubes shall be transfered to a clear glass container and after standing for 3 hours shall be examined for foreign particles visible to the naked eye. The lot size shall be expressed in units of tubes. The sample unit shall be one tube. The inspection level shall be S-2. If the sample average net contents per tube is less than specified in 5.1.1 or if there are visible foreign particles in the contents, the lot shall be rejected.

4.2.4 End item label testing. The end items (filled, sealed, and labeled tubes of insect repellent) shall be tested for the characteristics listed in table V. The lot size shall be expressed in tubes. The sample unit shall be one tube. Any test failure shall be cause for rejection of the lot. The sample size shall be in accordance with the following:

| Lot size (tubes) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and up | 5 |



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TABLE V. End item label tests

| Characteristic | Requirement | Test method |
|---|---|---|
| Label resistance to water | 3.3 | 4.3.9.1 |
| Label resistance to insect repellent lotion | 3.3 | 4.3.9.2 |

4.2.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

| Examine | Defect |
|---|---|
| Marking (exterior and interior) | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |
| Materials | Any component missing, damaged, or not as specified |
| Workmanship | Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling<br>Bulged or distorted container |
| Content | Number of tubes per container is more or less than required |

4.2.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

| Examine | Defect |
|---|---|
| Finished dimensions | Length, width, or height exceeds specified maximum requirement |



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| Examine | Defect |
|---|---|
| Palletization | Pallet pattern not as specified<br>Load not bonded as specified |
| Weight | Exceeds maximum load limits |
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |

4.3 Method of inspection.

4.3.1 Determination of DEET.

4.3.1.1 Equipment and GC conditions.

Equipment: A Hewlett Packard 5880A Gas Chromatograph (GC) or equivalent, equipped with Level Four Temperature Programming, Integrator and Methods, an automatic sampler and a 24 m x 0.3 mm I.D. SE-54 silicone capillary column.

GC conditions:
Gas Flow Rates: Auxiliary - 29 cc/min; Hydrogen - 30 cc/min;
Air - 428 cc/min; Split Vent - 60 cc/min
Injector 300°C
Detector 300°C (flame ionization)
Temp program - 10 min at 140°C, increase to 250°C at 25°C/min and hold for 5 min. Re-equilibrate at 140°C for 3 min. (This "burn off" may be omitted on standards.)
Injection - 1 ul splitless mode (V1=1) for first 0.3 min then open split valve (V1=0) for remainder of run.

4.3.1.2 Reference standard (see 6.3). Two, N,N-diethyl-m-toluamide (b.p. 111°C at 1 mm (ref. index nD25 1.5206)) (DEET) standards containing approximately 0.1781 and 0.2256 percent, by weight, of m-DEET in methyl ethyl ketone (MEK) are prepared. The exact percent of the meta(-m-)-isomer in the DEET used to make the standards can be determined using the Gas Chromatograph. The percent m-DEET in each standard is calculated using the following equation:

$$\text{Percent m-DEET standard} = \frac{\text{Weight DEET}}{\text{Total weight}} \times \text{Percent m-DEET}$$

where: Percent m-DEET standard is the percent of meta DEET in the standard.
Weight DEET is the amount of DEET used.
Total weight is the total weight of the DEET and MEK.
Percent m-DEET is the percent of meta-isomer in the DEET used to make the standard.



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4.3.1.3 Sample preparation. The samples to be analyzed for DEET are prepared by accurately weighing approximately 0.5000 g of each formulation into 80.5000 g (100.0 mL) of methyl ethyl ketone. These are shaken to dissolve/disperse the formulation. Approximately 2 grams of anhydrous magnesium sulfate are added to dry the solutions and then they are filtered.

4.3.1.4 Gas chromatographic analysis. A calibration table for m-DEET is set up as follows: The standard DEET solutions are run using the attached chromatographic conditions. The GC's method program will divide the calculated percent of m-DEET, which had been entered into the GC separately, for each standard, by their respective chromatographic areas. In this manner an amount/area correlation is set up for m-DEET. Next the diluted formulation samples are run. For each sample the GC will integrate the m-DEET peak area, compare this to the calibration table and then determine a percent m-DEET in the sample run on the GC.

4.3.1.5 Calculations. The percent m-DEET in the original formulation is calculated using the following equation:

$$\text{Percent m-DEET in formulation} = \frac{\text{Percent m-DEET GC} \times \text{Total weight}}{\text{Weight of formulation}}$$

where: Percent m-DEET GC is the Percent m-DEET for the diluted formulation determined by the gas chromatograph.
Total weight is the total weight of the diluted formulation = weight of formulation plus methyl ethyl ketone.
Weight of formulation is the weight of the formulation dissolved in the 100 mL of methyl ethyl ketone.

4.3.2 Density.

4.3.2.1 Apparatus.

a. 25-mL wide mouth pycnometer and stopper
b. Thermometer, Celsius

4.3.2.2 Procedure. Carefully fill a standardized wide mouth pycnometer with the sample that has been previously adjusted to 25°C by packing it firmly with the aid of a Teflon® spatula to remove any entrapped air. Press the stopper into the pycnometer and wipe off any excess material. Weigh the filled pycnometer.

4.3.2.3 Calculation.

$$\frac{\text{Weight of filled pycnometer(g)} - \text{tare weight of pycnometer (g)}}{\text{Weight of water (g) at 25°C}} = \text{Specific gravity @ 25°C}$$



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NOTE: If the specific gravity of the sample is to be determined at an elevated temperature, use the following formula:

$$G^1 = G - 0.00064\ (T - 25^oC)$$

where: $G^1$ = specific gravity at elevated temperature
$G$ = specific gravity at 25°C
$T$ = elevated temperature, °C

Density = Specific gravity at 25°C x 0.997 g/mL

4.3.3 Acid number. The acid number shall be determined in accordance with ASTM D 1639.

4.3.4 Viscosity.

4.3.4.1 Apparatus.

a. Suitably sized wide mouth jars with screw caps.

b. Constant temperature bath, 25° ± 1°C.

c. Viscometer, Brookfield Synchro-Lectric, Model LVF, LVT, or RVT.

4.3.4.2 Procedure.

a. Sample preparation.

(1) Fill sample jar approximately 3/4 full.

(2) Cap the jar and place in the constant temperature bath at 25° ± 1°C for 2 hours.

b. Test procedure Model LVF or LVT with spindle.

(1) With motor running, rotate speed control knob to the rpm specified in control procedure for product to be tested. Turn motor off.

(2) Attach spindle to lower shaft. Lift shaft slightly with one hand, avoiding side thrust, and screw spindle on with other hand. All spindles have a left-handed thread.



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(3) Insert spindle in the test material, centered horizontally in the container, until the fluid level is at the immersion groove cut in the spindle's shaft. Take care not to trap bubbles on the disc. Spindle must not be allowed to hit container sidewall while it is attached to viscometer, since this can damage shaft alignment.

(4) Level viscometer, using leveling screws on the base. The bubble level can be used to indicate accurate alignment.

(5) Depress clutch and turn on viscometer. Release clutch and allow dial to rotate until pointer position stabilizes or, in the case of thixotropic materials, through intervals of five dial revolutions. Thixotropic materials will be identified in individual control procedures.

(6) To take reading, depress clutch and snap motor switch to stop instrument with pointer in view. Record dial reading.

(7) Take two check readings. Start viscometer with clutch still depressed, then release and obtain readings.

(8) Remove spindle from viscometer with same care used in attaching it, and clean immediately.

c. Test procedure Model RVT or LVT with Helipath stand.

(1) Select bar-type spindle specified in specification and assemble by inserting top of spindle shaft through weight and securing into the chuck and closer coupling assembly.

(2) Attach assembled spindle to viscometer.

(3) Release height latch and adjust spindle so that crosspiece is covered by about 1/4 inch of material and is centered horizontally in containers.

(4) Level viscometer, using leveling screws on the base. Use bubble level to indicate accurate alignment.

(5) Start viscometer and allow dial to make two revolutions before turning on Helipath stand.

(6) Turn on Helipath stand and record each reading as needle passes midpoint of the dial windows. In this way, a figure will be obtained for the consistency at equal increments of drop through the material.

(7) Turn viscometer off after 1 inch of travel.

(8) Remove spindle from viscometer and clean immediately.



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4.3.4.3 Calculations.

a. Spindle.

(1) Average the three dial readings.

(2) Using Brookfield Factor Finder, locate correct factor for model, rpm, and spindle used.

(3) Multiply average dial reading by factor to obtain viscosity in cps.

(4) Record viscosity in cps, sample temperature, and model, rpm, and spindle used for the determination.

b. Helipath.

(1) Average the dial readings.

(2) Consulting "Helipath Stand Spindle Ranges" as specified in the owner's manual, locate the viscosity range for the model and the speed-spindle combination used.

(3) To determine viscosity in cps, multiply average dial reading by 0.01 of the range determined.

(4) Record viscosity in cps, sample temperature, and model, speed, and spindle used for the determination.

4.3.5 Fill volume of insect repellent.

4.3.5.1 Apparatus. Laboratory balance of sufficient gross capacity and at least 0.1 g accuracy.

4.3.5.2 Procedure.

a. Weigh the filled tube to the nearest 0.1 g, and record as W1. Record and report this volume as fill weight.

b. Empty the tube, wash and rinse it clean with water, and dry.

c. Weigh the empty tube, to the nearest 0.1 g, and record as W2.

d. Calculate the fill volume from the appropriate equation, record and report.

(1) Fill volume (mL per tube) = $\frac{(W1 - W2)}{(d)}$.



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$$(2)\ \text{Fill volume (fluid oz per tube)} = \frac{(W1 - W2).}{(29.57)\ (d)}$$

where: W1 = weight of the filled tube (g)
W2 = weight of the empty tube (g)
d = density of the liquid (g/mL) (see 4.3.2)
29.57 = conversion factor, mL to ounces (mL/oz).

4.3.6 Odor. Note presence of odor other than diethyltoluamide after saturating cloth or paper.

4.3.7 Microbial plate count.

4.3.7.1 Apparatus.

a. Sterile syringes, appropriate sizes
b. Sterile pipets, appropriate sizes
c. Petri dishes, 100 x 15mm, sterile
d. Auto clave, steam
e. Incubator, 33-37°C
f. Quebec colony counter
g. Hand-held tally counter

4.3.7.2 Media and reagents.

a. 99 mL sterile USP dilution blank

b. Soybean - Casein Digest Agar, USP (SCD) with lecithin and polysorbate 80

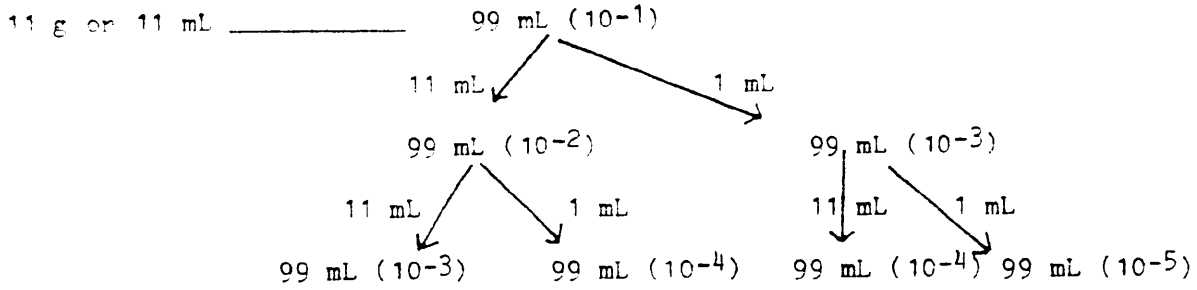
c. Standard Methods agar (SMA)

4.3.7.3 Procedure. Weigh 11 g or pipet 11 mL into a 99 mL sterile USP dilution blank. Heavy liquids can either be poured from their container or pipeted. The pipets are used to add samples by weight basis and not by volume. When weighing the sample, pipetting is preferred to pouring because a more accurate weight is achieved when adding by drops. Eleven grams into 99 mL gives a $10^{-1}$ dilution. Shake all dilution blanks with a one foot arc, 25 times in under 7 seconds. Pipet 1 mL of sample of $10^{-1}$ diluent into two petri dishes. Pour 15 to 20 mL of an appropriate medium (TSA-Tween SMA) tempered 44-46°C. Swirl plates gently to distribute the sample evenly throughout the medium. Permit the agar to solidify, invert the plates, and incubate at 35 ± 2°C for 48 hours.



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For samples expected to give higher counts, dilute to get plates containing 30 to 300 colonies.




11 g or 11 mL ---- 99 mL yields a dilution 10 times less concentrated
1 g or 1 mL ---- 99 mL yields a dilution 100 times less concentrated

4.3.7.4 Calculations.

a. Counting plates and recording results.

(1) Count plates having from 30 to 300 colonies.

(2) If a serial dilution ($10^{-1}$, $10^{-2}$, $10^{-3}$, etc.) was run, count only those plates yielding 30 to 300 colonies per plate. Multiply the number of colonies by the reciprocal of dilution and get an arithmetic average. If only one plate from the serial dilution falls from 30 to 300 colonies, count only that plate. If the higher dilution yields a count twice that of the lower dilution, count only the lower dilution.

(3) If all plates yield less than 30 to 300 colonies, count the total numbers directly and multiply by the reciprocal of the dilution.

(4) If all plates contain no colonies at all, express the results as less than "n" ("n" is the lowest dilution tested).

EXAMPLES: If a sample is diluted $10^{-1}$ and $10^{-2}$ and no colonies appear, express the result as 10/g ($10^{-1}$ was the lowest dilution). If a sample is diluted $10^{0}$, $10^{1}$, $10^{2}$ with no colonies appearing, express the result as 0, because $10^{0}$ was the lowest dilution.

(5) For plates containing more than 300 colonies per plate, get an average number per sq cm: for petri dishes 88 mm in diameter, using the equation for an area of a circle ($\pi r^2$) yields 60.82 sq cm or 61 sq cm rounded off.



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b. Discussion. For plates having more than 5 and less than 10 colonies per sq cm, count the total number of colonies in 7 representative squares (4 across and 3 vertical, or 3 across and 4 vertical) and divide the total number by 7 to get an average per sq cm. Multiply this number by 61 and by the dilution to get the number of bacteria per g or mL of the sample.

For plates having more than 10 colonies but less than 100 sq cm, get an average of colonies from 4 representative squares. Multiply this by 61 and the dilution factor.

For plates having greater than 100 colonies per sq cm, express the result as greater than 100 x 61 x dilution factor.

c. Summary.

(1) Plates containing less than 30 colonies:

| DILUTION | COLONIES | RESULTS |
|---|---|---|
| $10^{-1}$ | 0 | 10 |
| $10^{-2}$ | 0 | 100 |
| $10^{-1}$ | 2 | 20 |

(2) Plates containing from 30 to 300 colonies:

| DILUTION | COLONIES | RESULTS |
|---|---|---|
| $10^{-1}$ | 35 | 350 |

(3) Plates containing more than 300 colonies:

| DILUTION | COLONIES | RESULTS |
|---|---|---|
| $10^{-1}$ | $> 5, < 10$/cm | $n_{7/7} \times 61 \times 10$ |
| $10^{-1}$ | $> 10, < 100$/cm | $n_{4/4} \times 61 \times 10$ |
| $10^{-1}$ | $> 100$ | $> 100 \times 61 \times 10$ |

where: 61 = Surface area in petri dish
n = Number of colonies per 100 sq cm
$n_7$ or $n_4$ = Summation of colonies in 7 or 4 squares

4.3.8 pH determination.

4.3.8.1 Apparatus. Suitable potentiometric (pH) meter; pH combination electrode (hydrogen ion; calomel reference or equivalent).



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4.3.8.2 Reagents. USP standard buffer solutions.

4.3.8.3 Procedure.

a. Confirm that the electrode is properly filled.

b. Rinse the electrode with water.

c. Choose two standard buffers which most closely bracket the anticipated pH of the sample solution.

d. Immerse the electrode in one of the standard buffer solutions and calibrate the meter.

e. Rinse the electrode with water.

f. Check the standardization of the meter with the second buffer. Recalibrate the meter if the pH is more than ±0.07 pH unit of the listed value.

g. Rinse the electrode with water.

h. Immerse the electrode in a portion of the sample solution and, after 1 minute, record the pH.

i. Remove the electrode (do not rinse) and repeat step h until successive pH readings are reproducible to at least ± 0.05 unit. Wipe and rinse electrode if sample is not a low viscosity liquid before repeating step h.

j. Report the average of two successive pH readings reproducible to at least ±0.05 unit.

4.3.9 Labels.

4.3.9.1 Resistance to water. The tube shall be completely immersed in water at 70° to 80°F for one hour. Remove the tube and rub fingers over the face of the label to determine if the label either smears or loosens to the point that it can be moved. Any smearing or loosening of the label constitutes a failure.

4.3.9.2 Resistance to insect repellent. The tube shall be completely immersed in the type and concentration of repellent contained therein at 70° to 80°F for 1 hour. Remove the tube and rub fingers over the face of the label to determine if the label either smears or loosens to the point that it can be moved. Any smearing or loosening of the label constitutes a failure.



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5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A preservation. Four fluid ounces of insect repellent shall be packaged in a plastic tube. The tube shall be constructed of high density polyethylene and shall have a minimum thickness of 0.01 inch. Each tube shall conform to Olive Drab Color No. 24087 of FED-STD-595. The design and dimensions of the tube shall be as specified in figure 1. The cap shall be a 24 mm/410 design constructed of low density polyethylene and shall be black. The cap shall have a flip top dispensing closure that does not make a snapping noise when opened or closed. The orifice shall have a nominal size of 3.0 by 6.0 mm.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. One hundred twenty tubes, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC, grade V2s of PPP-B-636. Inside dimensions of the box shall be approximately 24 inches in length, 16 inches in width, and 6-1/2 inches in depth. Each box shall be fitted with partitions and top and bottom pad constructed from grade 275 corrugated fiberboard. Each box shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. One hundred twenty tubes, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Inside dimensions of the box shall be approximately 24 inches in length, 16 inches in width, and 6-1/4 inches in depth. Each box shall be fitted with partitions, and top and bottom pads constructed of grade 200 corrugated fiberboard. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. One hundred twenty tubes, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), tubes packed as specified in 5.2.2 or 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be



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bonded with straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Label. The label information shall be legibly and neatly printed, silk-screened directly on the tube. As an alternative, the label information shall be legibly and neatly printed on paper stock conforming to Olive Drab Color No. 24087 of FED-STD-595. The label shall be sealed to the tube with water-resistant adhesive. In all cases, the ink used shall be black, regardless of how the label information is applied to the tube. The label shall not smear nor be loosened from the tube when tested as specified in 4.3.9.

5.4.2 Labeling information. Labeling shall be in compliance with the requirements of the Federal Insecticide, Fungicide and Rodenticide Act as amended. Labeling shall contain, as a minimum, the directions for use and precautionary information as shown below. Labels shall be approved by the EPA.

5.4.2.1 Front label.

6840-01-284-3982
INSECT/ARTHROPOD REPELLENT LOTION
Military Specification MIL-I-44415
Contents: 4 Fluid Ounces (118 mL)

Repels mosquitoes, ticks, biting flies, chiggers, deer flies, fleas, and stable flies. Also repels terrestrial leeches in tropical areas where pest occurs.

Provides 95% or greater protection against mosquitoes for 12 or more hours under normal use conditions.

ACTIVE INGREDIENTS:

| | |
|---|---|
| N,N-diethyl-m-toluamide | 31.58% |
| Other isomers | 1.75% |
| INERT INGREDIENTS | 66.67% |

PRECAUTIONARY STATEMENTS:
Warning - For external use only. Keep out of reach of children. Causes eye irritation. Do not get into eyes. Do not apply to lips or excessively sunburned or damaged skin.

STATEMENT OF PRACTICAL TREATMENT:
If In Eyes: Flush with plenty of water. Get medical attention if irritation persists.

Contract No. DLA XXX-XX-X-XXXX



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5.4.2.2 Back label.

DIRECTIONS FOR USE:
It is a violation of Federal Law to use this product in a manner inconsistent with its labeling.

SKIN APPLICATION:
Squeeze into one hand approximately 2.5 mL of repellent a strip equal in length and width to the diagram on the side of the tube. Rub hands together and apply thoroughly in a thin layer to both forearms. Use additional lotion for upper arms. Repeat for other exposed areas. To apply to face, squeeze lotion into palm of hand and spread on face and neck. Avoid contact with eyes and lips. Repeat as necessary. Wipe hands after application.

May damage certain synthetic fabrics, plastics, painted or varnished surfaces. Avoid smearing on plastic eyeglass frames, goggles, watch crystals, etc. WILL NOT DAMAGE nylon, cotton, or wool fabrics.

Disposal: Do not reuse empty container. Wrap container and put in trash.

Manufacturer's Name and Address
Date of Pack and Lot Number

EPA Registration No. 58007-1
EPA Est. No. XXXXX-XX-XX

5.4.2.3 Side label. On side of tube, a box 2 1/2 inches in length by 3/8 inches in width, shall be printed. Inside the box, the following statement shall be printed: 2.5 mL strip diagram.

5.4.3 Registration of products. The products shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act and be in compliance with its requirements.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The repellent is intended for direct application on exposed skin surfaces of the body to protect against mosquitoes, biting flies (to include black, deer, horn, horse, snipe, greenhead, and stable flies), gnats, ticks, fleas, and chiggers (mites).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.



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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Level of packing required (see 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 Standard sample. A standard sample for the N, N-diethyl-m-toluamide in 4.3.1.2 can be obtained from either of the following:

a. Chemical Services Inc.
P.O. Box 3108
West Chester, PA 19381-3108

b. Aldrich Chemical Company, Inc.
Milwaukee, WI 53233

6.4 Trade names and suggested suppliers. Trade names and suggested suppliers for the chemicals listed in table I are shown in table VI. Chemicals from these companies will meet the stability requirements for the insect repellent in paragraph 3.3.

TABLE VI. Chemical trade names and suppliers

| Chemical | Trade name | Supplier |
|---|---|---|
| Purified water, USP | -- | -- |
| Mixture of 25% IP-III polymer in DEET | 3M, 41-4202-1922-6 | 3M |
| DEET | MGK diethyltoluamide 95% meta isomer min. | Hoechst Celanese Corp. |
| Propylene glycol dicaprylate/dicaprylate | Lexol PG 865 | Inolex Chemical Division of America |
| Polyethylene glycol (7) glyceryl ether | Liponic EG-7 | Lipo Chemicals, Inc. |
| Propylene glycol (15) stearyl ether | Arlamol E | ICI Americas, Inc. |
| Cetyl stearyl alcohol | ADOL 63 | Sherex Chemical Co. |



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TABLE VI. Chemical trade names and suppliers (cont'd)

| Chemical | Trade name | Supplier |
|---|---|---|
| Polyethylene glycol (82) glyceryl monotallowate | Varonic LI 48 | Sherex Chemical Co. |
| Amorphous silicon dioxide | CAB-O-SIL-M-5 | Cabot Corp. Cab-O-Sil Div. |
| Glyceryl stearate/sodium lauryl sulfate | Lexemul AS | Inolex Chemical Division of America |
| Polyethylene glycol (8) | Carbowax 400 | Union Carbide Corp. |
| Magnesium aluminum silicate | Veegum | R.T. Vanderbilt Co., Inc. |
| Polyethylene glycol (200) glyceryl monotallowate | Varonic LI 420 | Sherex Chemical Co. |
| Cetyl palmitate | Waxenol 816 | Wickhen Products, Inc. |
| Cellulose, 2-hydroxyethyl ether | Natrasol 250 HR | Hercules, Inc., Customer service |
| Diazolidinyl: urea: methyl paraben: propyl paraben: propylene glycol | Germaben II | Sutton Laboratories |

6.5 Subject term (key word) listing.

Bites, bug
DEET
Flies
Lotion
Mosquitoes
Protection, skin
Ticks



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MILITARY INTERESTS:

Custodians

Army - GL
Navy - SH
Air Force - 68

Review activities:

Army - MD
DLA - GS
AFPMB-PM

User activities:

Navy - MC, MS

COORDINATING ACTIVITIES CIVIL AGENCY:

USDA - AFS
GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

(Project 6840-1005)



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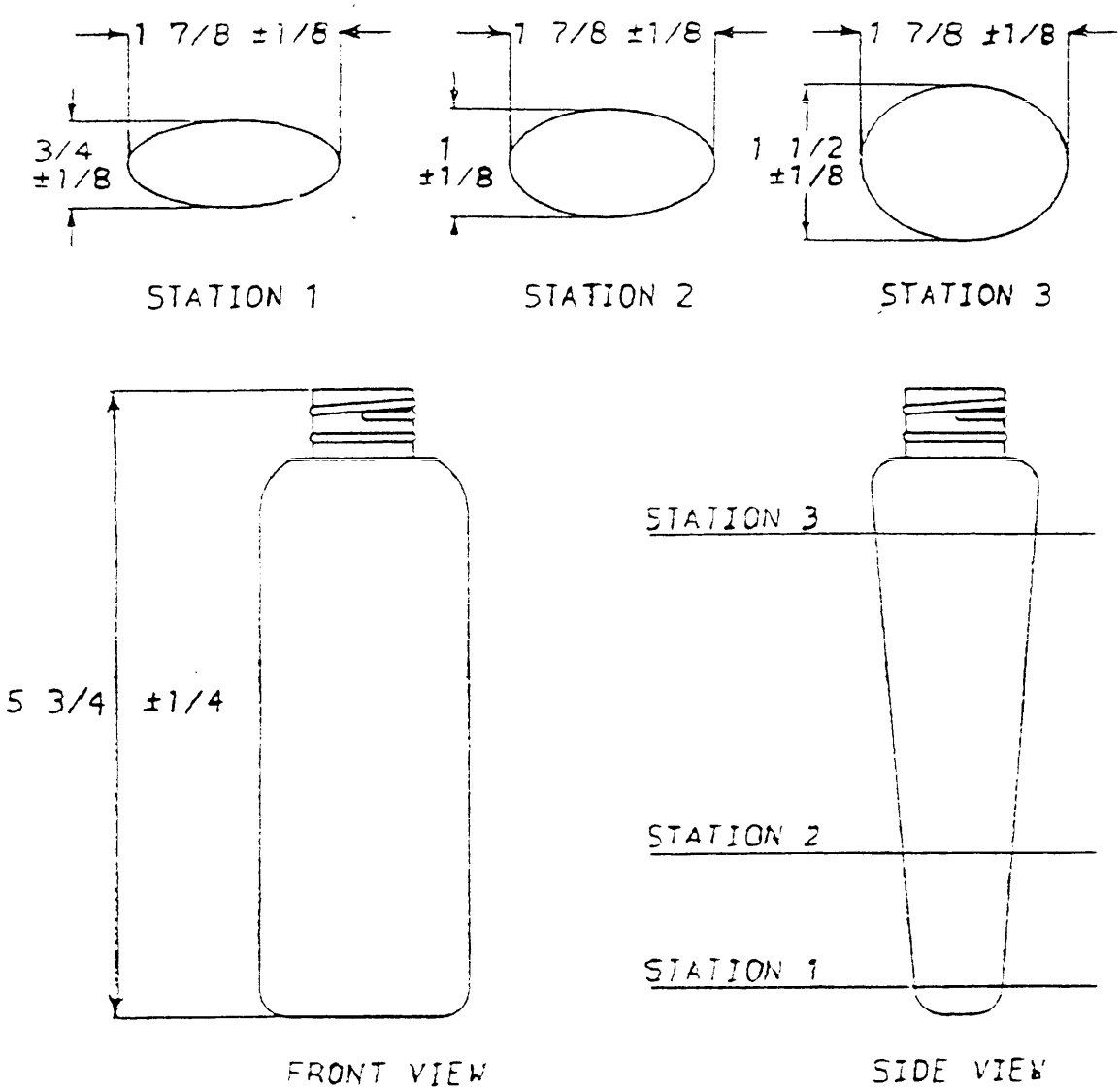



FIGURE 1. Tube design (4 oz. tube).



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-I-44415 | 2. DOCUMENT DATE (YYMMDD)<br>1990 December 21 |
|---|---|---|

3. DOCUMENT TITLE

Insect Repellent, Personal Application (Cream Lotion, Tube)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION |
|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) |
| | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON |
|---|---|---|
| U.S. Army Natick RD&E Center | 508-651-4532 | 256-4532 |
| c. ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: | |

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introduce them separately into blocks 1, 2 and 3. Send the contents to